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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	FAA!
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 8 (Decent Work and Economic Growth) & 4 (Quality Education)
Project / Solution Name	" UNHIRED" PROJECT
Name (Leader)	AHMAD IRFAN BIN HISHAMUDDIN
Email & Phone (Leader)	ahmadhshmddn@gmail.com / +60 12-853 4116
University Affiliation(s)	UNIVERSITI TEKNOLOGI MARA (UiTM)
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

The Malaysian workforce is experiencing a rapid technological shift, with 56% of professionals using AI at work, yet only 26% receiving formal training (Hays Asia Salary Guide 2025). This massive skills gap leaves workers uncertain about which competencies they need to remain relevant. This affects early-to-mid career professionals who risk technological displacement if they cannot adapt. Addressing this matters economically and socially; without structured upskilling pathways, we face rising structural unemployment and a widening inequality gap. Un-HiReD tackles this uncertainty by providing clear, data-backed guidance on exactly which skills individuals must build next to secure future-ready roles.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Un-HiReD is an AI career co-pilot designed to transition workers from their current skill sets to practical, future-ready roles. Unlike standard CV-to-chatbot tools, it begins with a user's specific goals, experience, and current skills. Utilizing the Groq API (**`openai/gpt-oss-20b`**), the application analyzes the user's profile against market trends derived from our local dataset (**`job_trends.csv`**). It connects existing skills to potential future roles, calculates a transparent skills-overlap match score, and clearly identifies knowledge gaps. Un-HiReD then translates these gaps into an actionable, ordered learning pathway, complete with estimated timelines and direct links to educational resources. The solution models potential career trajectories against different future scenarios and highlights the resulting SDG impact. Leveraging skills from Phase 1, we implemented a Next.js frontend with secure client-side storage (IndexedDB/localStorage) and a FastAPI Python backend hosted on Replit, ensuring rapid deployment, seamless GenAI API integration, and strict data privacy with in-memory processing.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js, React (Client-side storage via IndexedDB/localStorage)

Backend: FastAPI, Python 3, Pandas

AI Model/API: Groq API (**`openai/gpt-oss-20b`**)

Database/Data: Local `job_trends.csv`

Deployment: Replit

System Flow: User Interface (Next.js) → Backend (FastAPI, In-Memory Processing) → AI Inference (Groq API + `job_trends.csv` context) → Response rendered on UI.

Sponsor Resources Used: Yes. Replit is utilized for full-stack cloud hosting and live prototype deployment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking by empathizing with workers overwhelmed by AI disruption. We defined the core problem as a lack of actionable upskilling pathways rather than just a lack of jobs. Ideation led to a transparent matching system that avoids 'black box' anxiety by explicitly showing users their skill gaps and how scores are calculated. For prototyping, we prioritized user privacy and low friction; making CV uploads optional and processing data entirely client-side or in-memory builds immediate trust. Our deployment strategy utilizes Replit for highly accessible, zero-configuration cloud hosting, ensuring the prototype is robust for Phase 2 testing. We mitigated hallucination risks by grounding AI recommendations in a deterministic local dataset, ensuring the platform remains a reliable, user-centric tool.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 8.5 "Achieve full and productive employment and decent work for all women and men," and SDG Target 4.4 "Increase the number of youth and adults who have relevant skills for employment.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Only 26% of Malaysian professionals using AI have received formal training, leaving a 74% training deficit in the tech-adopting workforce.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Number of specific skill gaps identified and successfully mapped to actionable learning pathways per user profile.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	100% of identified skill gaps mapped to verified, free or low-cost learning resources for all tested user scenarios.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed**.**

Name	Type (code/dataset/model)	License
Next.js	Framework	MIT License
FastAPI	Framework	MIT License
Pandas	Code Library	BSD 3-Clause
openai/gpt-oss-20b	Groq (API)	Groq ToS / Open-source weights

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



Name: Ahmad Irfan bin Hishamuddin (Group Leader)

[Date]
24/10/2026